



Hyper-Local Crop Advisories with **ES|QL**, **ELSER**, and **Elastic Agent Builder**

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The moment that started this

“ *"My maize leaves are turning yellow and curling. We've had almost no rain for 3 weeks. The plants are young. **What's wrong, and what do I do – today?**"* ”

A smallholder farmer, with a phone, needs an answer **now**.

Where do they get it?

The Problem

Smallholder farms (<2 ha) are **~84% of the world's farms** and grow roughly **a third of our food** — yet:

- 🗺️ **Generic advice** — manuals don't know *your* soil, *your* recent rain, *your* local pests
- 🧑🌾 **No agronomist** — expert access is scarce, slow, expensive
- 📋 **Clunky tools** — forms and dashboards, not plain language
- ⚙️ **No integration** — nothing answers "*given my exact rainfall + soil + nearby outbreaks, why are my leaves curling?*"

Why that question is hard

Answering it means combining **four independent signals**, fast:

Signal	Example
 Recent local climate	rainfall 60% below normal, last 90 days, within 200 km
 Regional pest outbreaks	Fall Armyworm reported 180 km away, HIGH severity
 Soil reality	Ferric Lixisol, well-drained → loses moisture fast
 Agronomic knowledge	"leaves curling + low rain at 6-leaf" → drought stress

No single off-the-shelf tool does **geo + time-series + semantic + reasoning** together.

The Solution: FarmSense

An **AI agronomist in your pocket** — free, instant, hyper-local.

1. Farmer describes the situation in **plain language** (Kibana Chat **or** Telegram)
2. An **LLM agent** reasons across the **four data streams** in Elasticsearch
3. Returns a concrete advisory: **risk → diagnosis → actions → what to watch**
4. Every advisory feeds a **regional early-warning map** + audit/alert workflow

What the farmer gets back

RISK LEVEL: ● HIGH

PRIMARY DIAGNOSIS: Drought stress at the 6-leaf stage.

WHY: Rainfall ~10–12 mm/week (60–65% below normal); temps 31–35°C;
Ferric Lixisol loam drains fast – moisture isn't retained.

IMMEDIATE (48h): mulch today; water at root zone if any source.

PREVENTIVE (2 wks): weed by hand; split nitrogen when rain returns.

WATCH FOR: tight leaf rolling / brown tips → escalate watering.

Plain language. Specific numbers. **No expensive inputs assumed.**

Why Elasticsearch?

(and not the usual 6-tool stack)

The naive architecture: 6 systems

To do geo + time-series + semantic + agent + alerts + dashboards, teams usually wire together:

PostGIS + **InfluxDB** + **Pinecone** + an embedding service + **LangChain** + **cron/back-end** + **Tableau**

➡ Six systems. Six security models. Lots of glue code. Data copied everywhere.

FarmSense: one platform

Need	Typical separate tool	FarmSense uses Elastic
Geo-distance / spatial	PostGIS / GIS DB	ES QL <code>ST_DISTANCE</code> on <code>geo_point</code>
Time-series rollups	InfluxDB / Timescale	ES QL <code>BUCKET</code> over <code>@timestamp</code>
Semantic search	Pinecone + embeddings	ELSER v2 <code>semantic_text</code> (built-in)
Agent orchestration	LangChain / custom	Elastic Agent Builder
Automation / alerts	cron + backend	Elastic Workflows
Maps / dashboards	Tableau + map tiles	Kibana Maps + Lens (ES QL)

One datastore • one query language • one security model.

Geo + time-series in one ES|QL query

`geo_climate_query` — weekly rainfall within 200 km, last 90 days:

```
FROM climate-timeseries
| WHERE ST_DISTANCE(point, TO_GEOPOINT(?lat_lon)) < 200000
      AND ts >= NOW() - 90 days
| STATS avg_rainfall = AVG(rainfall_mm),
      avg_temp      = AVG(temp_max_celsius)
  BY week = BUCKET(ts, 1 week)
| SORT week DESC
```

Spatial filter + time bucket + aggregation — no joins, no second database.

Semantic search without the plumbing – ELSER

The farmer says *"leaves look weird."* The guidance says *"chlorosis."*

- `crop-knowledge.text_semantic` is a `semantic_text` field → ELSER v2 runs at **index & query time automatically**
- **No** embedding microservice, **no** model hosting, **no** vector-DB sync job
- Bridges the vocabulary gap farmers actually have

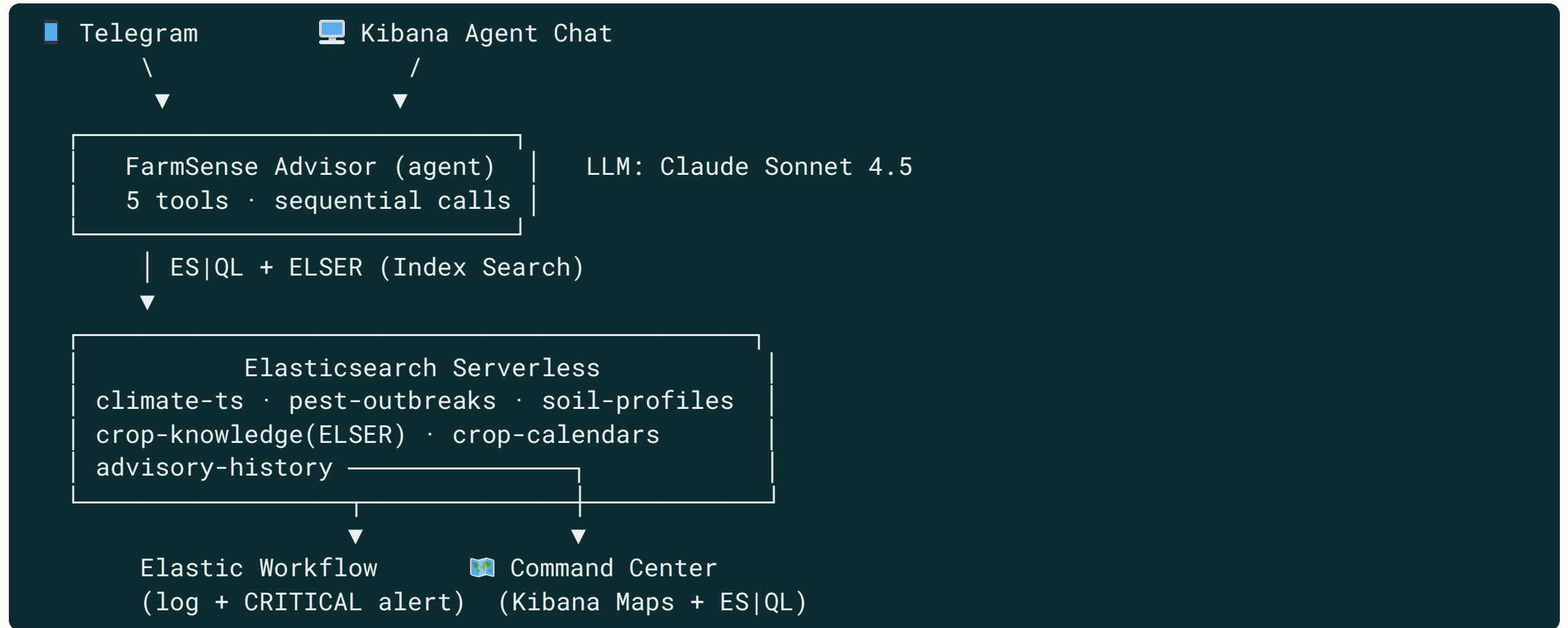
```
"text_semantic": { "type": "semantic_text",  
                  "inference_id": ".elser-2-elasticsearch" }
```

The agent: Elastic Agent Builder

- **FarmSense Advisor** orchestrates **5 tools** + **Claude Sonnet 4.5**
- Tools are **declarative**: 4 × ES|QL + 1 × Index Search (ELSER) — no orchestration code
- Native **multi-turn conversations** (chat *and* our Telegram bot)
- The agent queries the **same indices** Kibana maps — zero data movement

```
geo_normalize → geo_climate → pest_outbreak → crop_knowledge → soil_profile → advisory
```

Architecture



The 4-step pipeline

1. **Intake** — parse crop/location/symptoms → `geo_normalize` to lat/lon (WKT)
2. **Intelligence** — climate · pest · soil · knowledge (one tool at a time)
3. **Advisory** — synthesize: risk · diagnosis · actions · watch-for
4. **Log & Alert** — `advisory-alert-workflow` indexes it + webhook on CRITICAL

 End-to-end in **~40-70 seconds**.

The "wow": FarmSense Command Center

Every advisory is also a **data point**.

- Aggregate **advisory-history** → a **live regional pest & drought early-warning map**
- Kibana **Maps** (points colored by risk) + **ES|QL** panels: risk mix, top pests, crops at risk, trend
- **One dataset, two products**: 1:1 farmer help → population-scale intelligence for NGOs & ministries
- Built on indices we already have — **no new pipeline**

Architecture decisions & trade-offs

- **Sequential, not parallel, tool calls** — the chat UI sent *empty args* to parallel tool calls and looped; one-at-a-time made it reliable
- **Model matters** — the default model fumbled tool params; **Claude Sonnet 4.5** fills them perfectly (a required pick)
- **Workflow as a platform component** — workflow-type tools didn't expose inputs to the LLM, so logging/alerts run via the Workflows engine, not an agent call
- **Synthetic data, real schemas** — mirrors NASA POWER / ISRIC / FAO so it's production-swappable

Lessons from an Agent Builder preview

- 🧠 **Tool-call reliability is a prompt problem** — be explicit: one tool at a time, never empty args
- 🔌 **Pick your connector** — model choice is the difference between "works" and "loops"
- 🌐 **Reverse-engineer saved objects** — build one panel in the UI, export, clone via API
- 📈 **Seed your audit index** — dashboards need volume to look alive (we seeded 220 advisories)

What's next 🚀

- 📸 **Snap-a-Leaf** — photo of the leaf → **image embedding + kNN vector search** → disease match
- 📡 **Guardian Mode** — a **scheduled Workflow** pushes "FAW 40 km away — act now" *before* the farmer asks
- 🗣️ **Any-language voice** — voice note → transcribe → advise → reply in the farmer's language
- 🛰️ **Satellite NDVI** — detect crop stress from space before it's visible

Live Demo

Kibana Agent Chat → maize (HIGH) & rice (CRITICAL)

Telegram → multi-turn on a phone

Command Center → the regional map

Recap

- A real problem: **hyper-local crop advice**, in plain language, instantly
- **One Elastic platform** replaced a 6-tool stack: ES|QL · ELSER · Agent Builder · Workflows · Maps
- A working agent **and** a regional early-warning Command Center from the **same data**
- Honest trade-offs, real lessons, a clear roadmap

Thank you! 🌱 — Questions?

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github.com/ankitlade12/farmsense